TAKING THE P-TH POWER OF THE VARIABLE DOES NOT CHANGE THE NUMBER OF ROOTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The reduction step that lets the separable-degree machinery treat x^p-substituted polynomials, cited in the separable degree discussion above its definition.

Editorial note. The label lemma-nr-roots-unchanged names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The algebraic closure is algebraically closed: every element has a p-th root
- Freshman's dream exponent law in characteristic p: (alpha - beta)^p = alpha^p - beta^p, so equal p-th powers differ by a p-th power of an element of the field, which is zero only for zero

Lemma 0.1. *Let F be a field and let $\overline{F}$ be an algebraic closure of F. Let $p > 0$ be the characteristic of F. Let P be a polynomial over F. Then the set of roots of P and $P(x^p)$ in $\overline{F}$ have the same cardinality (not counting multiplicity).*

Proof. Clearly, α is a root of $P(x^p)$ if and only if α^p is a root of P. In other words, the roots of $P(x^p)$ are the roots of $x^p - \beta$, where β is a root of P. Thus it suffices to show that the map $\overline{F} \to \overline{F}$, $\alpha \mapsto \alpha^p$ is bijective. It is surjective, as $\overline{F}$ is algebraically closed which means that every element has a pth root. It is injective because $\alpha^p = \beta^p$ implies $(\alpha - \beta)^p = 0$ because the characteristic is p. And of course in a field $x^p = 0$ implies $x = 0$. $\square$

Date: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.